Math and Theory: collection overview

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Research process and AI disclosure

This collection reports fully autonomous agentic research. Dev D. Goyal is responsible for publication. Models used: GPT 6 Astra by OpenAI in ChatGPT and Codex, GPT 6.1 Sol by OpenAI in Codex, and Opus 5.5 by Anthropic used with Claude Code. Each paper records how they were used.

Opus 5.5 by Anthropic used with Claude Code assessed the October 2026 uploads on ADMM multiplier steps, Nesterov flows, Chambolle–Pock cycles, critical hard-core correlations, Schmidt numbers of the Tiles state, a binary spectral tile, the Keller–Klein conjecture, unit square classes of totally real fields, depth-one Artin–Schreier towers, the upper PPT entanglement-cost hierarchy, matrix-pencil rank distance, selfadhesivity of polymatroids, nonnegative rank of Kronecker squares and multi-marginal maximal monotonicity, coordinated independent AI referee checks, extended the planar Nesterov result, classified and formalized the complements of the binary tile, ran fresh Lean replays, wrote or revised their fourteen manuscripts, and updated this overview and the research website. The research in those uploads was produced by autonomous AI agents; apart from the hard-core, tiling and Keller–Klein packages, which record ChatGPT sessions, the unit-squareclass package, whose tool logs indicate OpenAI Codex, and the multi-marginal Minty package, which records an OpenAI GPT-6 Astra workflow, their records do not name the system used.

AI checks are not expert human validation. These preprints have not been externally peer reviewed, and the collection makes no claim of first priority. Correspondence received after publication brought earlier or overlapping work on six results to light, and a check against public problem records on 31 October 2026 found earlier or related public work for eight more; the priority and novelty record documents when it was learned and what remains uncertain. The authorship disclosure identifies the tools and their roles.

Abstract

The catalogue contains forty-five manuscripts and research notes across 44 result families in mathematics, theoretical computer science, and mathematical physics. Each paper states its formal verification scope. The thermal proof covers finite spectral statements, without a formal stochastic long-time identification theorem. The Gyires preprint proves inequalities for structured matrices and develops related polynomial bounds. Its principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have complete Lean proofs. The results comprise counterexamples, exact special cases, sharp sequence bounds, and convergence thresholds for optimization methods. The collection makes no claim of first priority. The papers have not been peer reviewed.

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1 How to use the collection

The manuscript catalogue links each paper to its formal statements and independent computations. The verification guide records the commands and their results. Seven families share a Lake project pinned to Lean 4.35.0. Its 29 selected Comparator statements include equality, sharpness, and auxiliary positivity claims. The decision-tree and thermal packages use separate Lean 4.19.0 environments and a pinned Mathlib revision. The decision-tree package has 29 axiom reports of its own. This is a separate count from the shared Comparator selection.

2 Mathematics

2.1 Matrix theory

2.1.1 Real positive-definite counterexamples to extended q-permanent monotonicity

For every real parameter $q > 1$, some real symmetric positive-definite matrix has a negative q-permanent derivative at q. An explicit matrix of order five decreases from $q = 14$ to $q = 15$. The construction also gives correlation matrices with no zero entries and integer-entry examples. The dimension can depend on q. The real-symmetric question on $[-1, 1]$ remains outside the result.

Scope: Counterexample and all-parameter extension.

Manuscript Formal statement and prior work.

2.1.2 Gyires's permanent inequality for structured matrices

Every nonnegative real circulant matrix A of order five satisfies $24\,\mathrm{per}(AA^T) \le 625\,\mathrm{per}(A)^2$. Equality holds exactly for constant matrices. This proves the order-five circulant case of Gyires's ratio inequality. A bound in sorted-gap coordinates has sharp coefficient 484. The conjecture for arbitrary doubly stochastic matrices remains open.

Scope: Structured matrix inequalities and related polynomial bounds. The principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have Lean proofs.

Manuscript Formal statement and prior work.

Broader arguments in the same preprint

The manuscript also proves results for circulants of orders six and seven, with sharp sorted-gap margins. It treats $aI + bQ + cQ^2$ in every dimension, where Q is any permutation matrix and the coefficients are nonnegative. Further results cover four-block doubly stochastic matrices with unequal block sizes and boundary entries, matrices with one exceptional column or row, and every rank-at-most-two doubly stochastic matrix through order fifteen.

The supporting sections prove structural, scalar, deletion, balancing, local polynomial, and configuration-model results. Exact counterexamples identify the limits of auxiliary assertions. The complete support record gives the source map and exact reproduction commands.

The all-order four-block and exceptional-vector theorems have complete Lean proofs. Their inequalities and equality cases are among the 29 selected formal statements. The rank-two theorem has a complete Lean proof for orders 2 through 8. The general order-seven circulant theorem and rank-two orders 9 through 15 remain incomplete in Lean. Separate Lean packages prove the principal order-six and all-order permutation-polynomial theorems. The all-order rank-two problem and the unrestricted Gyires conjecture remain unproved here.

2.1.3 A polygon enclosure fails for sums of normal matrices

A polygon enclosure fails for a specified pair of normal matrices and its complete transposition graph. Actual normal matrices, determinants, straight-segment exterior and component exclusion are formalized. Correspondence with an external source conjecture is conditional.

Manuscript Reproduction and source context.

2.1.4 An unbounded obstruction to projection AM–GM in Loewner order

Three rank-one orthogonal projections in dimension three violate the symmetrized-product Loewner AM–GM bound. No finite real multiplier repairs it, even with a positive-definite mean. The examples do not satisfy the additional nonsingular block structure of the motivating coordinate-descent application.

Formal scope: Lean verifies a rational three-projection matrix obstruction and a parameterized obstruction for every real multiplier in dimension three; rank-one, positive-definite mean, rational all-multiplier choices, and the standard minimal-factor observation have complete conventional proofs.

Manuscript Proof and reproduction.

2.1.5 Matrix-pencil rank distance as an explicit completion problem

Over every infinite field, two matrix pencils have strict-equivalence classes at normal-rank distance at most x exactly when an explicit dilation of the first is a constant full-rank compression of the dilation of the second plus x scalar blocks. With a cited block minimal bases theorem this

proves Conjecture 1 of Dodig and Stošić: the bounded rank perturbation problem for matrix pencils is a completion problem. The explicit theorem is formalized in Lean 4.35.0-rc3.

Manuscript Proof and reproduction.

2.1.6 Counterexamples to the Kronecker-square lower bound for nonnegative rank

For every $k \geq 0$ an explicit rational nonnegative matrix C of order $k+11$, rank $k+10$ and nonnegative rank $k+11$ satisfies $\operatorname{rank}_+(C \otimes C) \leq (k+10)^2 + 9$. This disproves the conjecture of Vandaele, Gillis, Glineur and Tuyttens that $\operatorname{rank}_+(M \otimes M) \geq \operatorname{rank}_+(M) \operatorname{rank}(M)$. The family is formalized in Lean 4.35.0-rc3.
Manuscript Proof and reproduction.

2.2 Graph theory and graph polynomials

2.2.1 An entropy increase for a structurally disassortative DomiRank graph

A connected graph with 14 vertices and 32 edges has negative degree–log-degree edge covariance. All resolvent scores are positive on $[251/1000, 63/250]$ inside the stable domain. Its entropy nevertheless increases by more than $1/10000$ in natural-logarithm units. This refutes the source's assertion for every positive-score interval.

Scope: Counterexample to an interval monotonicity conjecture.

Manuscript Formal statement and prior work.

2.2.2 Domination sequences of powers of paths and cycles

The domination counts of every natural power of a path or cycle form an ultra-log-concave sequence. Dividing each count by its binomial coefficient gives the normalized log-concavity inequality. The proof includes all natural graph orders and powers. It does not assert the same property for arbitrary graphs or real-rootedness of their domination polynomials.
Manuscript Formal proof and reproduction.

2.2.3 Sharp escape of complex domination roots in square flowers

Sharp complex-root escape for square flowers, with complete flower and book real-root classifications and multiplicities. All numbered manuscript conclusions are formalized in seventy modules with 258 endpoint audits. Historical novelty and exact external-source correspondence remain provisional.

Manuscript Reproduction and source context.

2.2.4 Leaf deletion can move the unique mode of a tree domination polynomial by two

Deleting a leaf can move the unique mode of an explicit tree domination polynomial by two. The fixed tree, deletion, actual domination counts and unique-mode movement are formalized. External question identity and historical priority remain provisional.

Manuscript Reproduction and source context.

2.3 Inequalities and real analysis

2.3.1 Diagonal concavity does not imply concavity for sums of elementary symmetric polynomials

For $M > 0$, let $f_M = 1 + M\sigma_1 + \sigma_2 + M\sigma_3 + \sigma_4$ in four variables, where σ_j is the elementary symmetric polynomial of degree j. The fourth root $f_M^{1/4}$ is concave on the positive diagonal

exactly when $M \geq 1$. It is concave on the full positive orthant exactly when $M = 1$. Thus every $M > 1$ disproves Lachaume's criterion with strictly positive coefficients.

Scope: Counterexample and complete parameter classification.

Manuscript Formal statement and prior work.

2.4 Integer sequences and combinatorics

2.4.1 Alternating Motzkin sums

The principal root bound and bounded-forcing companion form one sequence family.

Sharp root log-concavity of alternating Motzkin sums Let $T_n = \sum_{j=0}^{n} (-1)^{n-j} M_j$, where M_j is the Motzkin number. The sequence $T_n^{1/n}$ is strictly log-concave on its tail starting at five. The inequality reverses at center five, so no earlier tail is log-concave. This strengthens Zhao's proposed tail starting at nine. The source recurrence and finite sum are linked in Lean.

Scope: Conjecture resolution with an optimal threshold.

Manuscript Formal statement and prior work.

Sharp log-concavity for alternating Motzkin sums A robust bounded-forcing recurrence theorem and corrected scaled corollary, with applications to alternating Motzkin-type sums. The arbitrary-real recurrence theorem and scaled corollary, including zero forcing, are formalized. Concrete sequences, uniform-family applications, root transfer and threshold classification retain conventional proofs.

Manuscript Reproduction and source context.

2.4.2 Counterexamples to a late-iterate ordering conjecture for generalized Hofstadter functions

For every $k \geq 4$, some input n satisfies $F_k^{\circ(2k+1)}(n) > F_{k+1}^{\circ(2k+2)}(n)$. Here the superscript denotes repeated composition. The proof constructs inverse iterates with exact gap $k - 3$. The witness $F_4^{\circ 9}(132) = 8 > 7 = F_5^{\circ 10}(132)$ gives the first instance. The result concerns infinitely many parameters. It does not assert infinitely many violating inputs for each fixed parameter.

Scope: Counterexamples for every parameter k≥4.

Manuscript Formal statement and prior work.

2.4.3 A positive constant-recursive sequence with nonquasipolynomial power ranks

Exact minimal recurrences for every power of a rank-thirteen sequence, nonquasipolynomial ranks, and infinitely many power-rank profiles at each rank at least three. The main and arbitrary-rank companion theorems are formalized. Generating-function and polynomial corollaries retain conventional proofs.

Manuscript Reproduction and source context.

2.5 Quadratic forms and number theory

2.5.1 A rank bound for represented totally positive unit squareclasses

A totally positive definite quadratic lattice of rank n over a totally real algebraic field, with integral norms but not necessarily classical, represents at most n classes of totally positive units modulo squares of units. This answers Question 3.4 of Daans, Kala, Man, Widmer and Yatsyna negatively; the bound is a short consequence of standard facts on finite reflection groups. The theorem is formalized in Lean 4.19.0 for all algebraic totally real fields.

Manuscript Formal proof and reproduction.

2.6 Valuation theory

2.6.1 Depth-one Artin–Schreier defect towers are composita

Over a perfect henselian subfield K of $k((y^{\mathbb{Q}}))$ containing $k(y)$ and without finite accumulation points, a nontrivial finite Artin–Schreier tower of depth one is a compositum of Artin–Schreier extensions of K. This is the converse conjectured by Nart and Novacoski. Both implications are formalized in Lean, with depth one defined by Mac Lane–Vaquié chains in a Lean 4.35 project and expressed through Okutsu sequences in a Lean 4.19 project.
Manuscript Proof and reproduction.

2.7 Discrete analysis, tilings, probability and entropy

2.7.1 The sharp Fourier bound at density one eighth

For every indicator of mean $1/8$ on a finite uniform Boolean cube, the level-one Fourier weight is at most $3/64$. A codimension-three subcube attains this bound in every dimension at least three. The proof also gives the equivalent sharp average Hamming-distance bound. It uses exact rational polynomial certificates and a dimension-independent moment identity. Fang and Huang proved this bound earlier, together with the equality cases; no originality or priority is claimed for it.
Manuscript Formal proof and reproduction.

2.7.2 A 64-point spectral tile in $\mathbb{F}_2^9$ with no periodic complement

An explicit 64-point subset E of $\mathbb{F}_2^9$ is spectral and tiles by translation, but none of its tiling complements is periodic: they are exactly the 512 translates of one aperiodic eight-point set. Hence no linear subspace complements E, and $\mathbb{F}_2^9$ fails the periodic tiling property. This answers negatively the subspace-complement part of a question of Ferguson and Iosevich; whether every spectral 64-point subset of $\mathbb{F}_2^9$ tiles remains open. The main theorem and the classification of complements are formalized in Lean 4.19.0.
Manuscript Formal proof and reproduction.

2.7.3 The small-coefficient upper half of the Keller–Klein conjecture

For a Rademacher sum $X = \sum_i v_i \varepsilon_i$ with $\sum_i v_i^2 = 1$ and every $|v_i| \le a \le 7/100$, $\Pr(|X| \le x) \le 2\Phi(x) + 2\Phi(a) - 2$ for all $x \ge 0$. This closes the range of the Keller–Klein conjecture left open by Ivanisvili's computer-assisted manuscript of 21 October 2026; together with his results it implies the full conjecture, conditionally on those unrefereed results. The theorem is formalized in Lean 4.19.0.
Manuscript Formal proof and reproduction.

2.7.4 A quartic entropy deficit for random rankings

Let P be a probability distribution on the complete rankings of n labels. If every pairwise comparison has probability at distance at least $\varepsilon > 0$ from $1/2$, then

$$H(P) \le (1 - \varepsilon^4/144) \log(n!).$$

The formal theorem includes every finite size, including zero and one. It assumes neither independent comparisons nor transitive preferred directions. It covers both ranking lists and position-valued permutations.

Manuscript, Lean proof, and reproduction commands.

2.7.5 Sharp universal Rényi threshold for iid sums on finite Sidon sets

A sharp universal Renyi-order threshold for iid sums supported on finite strong-B2 sets. The finite-support iid theorem, zero probabilities and critical endpoint are formalized. Comparison with unverified source definitions is conditional.

Manuscript Reproduction and source context.

2.7.6 Structural selfadhesivity does not imply a numerically selfadhesive realization

An explicit five-variable conditional-independence model is structurally selfadhesive at every overlap, yet no real polymatroid inducing it is selfadhesive in Matúš's sense. This answers Question 4.18 of Boege negatively. Both parts are formalized in Lean 4.35.0-rc3 for arbitrary real polymatroids.

Manuscript Proof and reproduction.

2.8 Matroid theory

2.8.1 The asymmetric Merino–Welsh region

The theorem characterizes every nonnegative pair (x, y) for which

$$T_M(x, 0)T_M(0, y) \geq T_M(1, 1)^2$$

holds for all finite matroids without loops or coloops. The criterion uses two explicit scalar inequalities for every integer $k \geq 2$. The proof includes empty ground sets and finite ground sets in arbitrary ambient types. The package fetches Liu's pinned proof library from its public repository.

Manuscript Formal proof and reproduction.

2.9 Optimization

2.9.1 A narrow interval for the universal multiplier step of two-block ADMM

Two-block ADMM with relaxation only in the multiplier update converges for every constant step $0 < \tau \leq 943/500$ on finite-dimensional convex problems whose proximal subproblems attain their minima and which have a saddle point. Fixed compact polytopes in $\mathbb{R}^{128}$, with zero objectives, admit a nonconstant periodic orbit for every $\tau \geq 897744153596994489/475778439151265168 \approx 1.8868954$. The universal threshold therefore lies in an interval of width below 0.0009; its exact value is open. The convergence theorem, its affine-constraint versions, both finite-memory certificates and the periodic orbit are formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

2.9.2 Escaping Nesterov flows and iterates below critical damping

For the Nesterov differential equation $X'' + (r/t)X' + \nabla F(X) = 0$ with zero initial velocity, and for the Nesterov iteration with momentum $k/(k + r)$, universal boundedness over convex objectives with a 1-Lipschitz gradient and an attained minimum holds on the Hilbert space ℓ^2 exactly when $r \geq 3$. An explicit convex planar potential has an escaping trajectory for every $0 < r \leq 2$, and its trajectories from the same start are bounded for $r > 2$. The finite-dimensional range $2 < r < 3$ remains open. The Hilbert-space and discrete thresholds and the planar escape for $1 < r \leq 2$ are formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

2.9.3 Periodic obstructions to small-extrapolation Chambolle–Pock

For the unmodified Chambolle–Pock method, every step product $\kappa = \tau\sigma\|L\|^2 > 2$ admits fixed compact polytopes in $\mathbb{R}^4$ whose iteration is periodic and nonconvergent for all sufficiently small extrapolation $\theta \geq 0$. Any universal sufficient condition $\kappa < c(\theta)$ therefore has $\limsup_{\theta\downarrow 0} c(\theta) \leq 2$, below the quadratic bound $4/(1+2\theta)$. An exact universal three-cycle threshold and a rational seven-cycle are also proved. The constructions are formalized in Lean 4.19.0; the obstruction intervals in θ are very small.

Manuscript Proof and reproduction.

2.9.4 A rational five-cycle for two-block ADMM with multiplier step below two

An exact period-five orbit for multiplier-only two-block ADMM at step 199/100. Seven modules and eleven endpoint audits cover the orbit, minimizers and optimizer. Analytic regularity of the finite strongly convex extension retains a conventional proof.

Manuscript Reproduction and source context.

2.9.5 The multi-marginal converse of Minty's theorem fails in the plane

For three marginals on $\mathbb{R}^2$ with the cost $\sum_{i<j}\langle x_i, x_j\rangle$, an explicit three-point c-monotone set admits no compatible tuple with sum zero, so every maximally c-monotone extension has a sum image that omits the origin and is not convex. This answers negatively the question of Remark 2.6 of Bartz, Bauschke, Phan and Wang. The argument is formalized in Lean 4.19.0.
Manuscript Proof and reproduction.

2.9.6 The exact last-gradient bound for constant-step gradient descent in the strongly convex case

The exact final-gradient guarantee holds on every real Hilbert space for $\mathrm{L} > 0$, $0 \leq \mu \leq \mathrm{L}$, normalized step $0 \leq \mathrm{h} \leq 2$ and every natural horizon. Actual one-dimensional Huber and quadratic functions attain its two branches.

Formal scope: The actual-function upper bound and normalized actual-function attainment are formalized, including every horizon and closed parameter boundary. Sharpness for arbitrary $\mathrm{L} > 0$ and radius $\mathrm{R} > 0$ is proved by rescaling in the manuscript, without a separate Lean wrapper.

Manuscript Proof and reproduction.

3 Theoretical computer science

3.1 Sampling algorithms and information theory

3.1.1 Critical hard-core influence bounds via nonbacktracking collisions

For every $\Delta \geq 3$, every row of the hard-core influence matrix has absolute sum $O_\Delta(\sqrt{n}\log n)$ on every n-vertex graph of maximum degree at most Δ, at every activity up to the uniqueness threshold and under every feasible pinning, with an explicit but large constant. This recovers, up to a logarithm, an $O(\sqrt{n})$ bound stated in 2025 and withdrawn in 2026. The influence bound is formalized in Lean 4.19.0; the mixing-time consequence for Glauber dynamics depends on a published proposition and is not formalized.

Manuscript Proof and reproduction.

3.1.2 Counterexamples to a best-of-n win-rate bound at every budget

For every $n \geq 2$ and $\delta > 0$, a finite positive reference distribution with largest atom at most δ violates the proposed best-of-n divergence–win-rate curve. A four-response best-of-two example gives exact probabilities and a strict logarithmic certificate. The all-budget proof uses finite interval bounds and gives a quantitative positive gap. It does not determine the sharp replacement curve.

Scope: Disproof for every budget n≥2 and every positive atom cap.

Manuscript Formal statement and prior work.

3.2 Boolean functions and decision trees

3.2.1 Logarithmic influence bounds and variance counterexamples for Boolean decision-tree covariance

Let T be a finite Boolean decision tree under uniform independent bits. No coordinate repeats on a root-to-leaf path. Let $k_i(T)$ count all query nodes labelled i, and let $I_i(f)$ be the probability that flipping coordinate i changes the output. The weighted sum of absolute child covariances satisfies

$$\sum_v 2^{-d(v)} \left|\mathrm{Cov}(f_{v,+}, f_{v,-})\right| \leq \sum_i I_i(f) \log_2 \max\{1, k_i(T)\}.$$

Here $d(v)$ is the node depth, and the two child functions use the same input cube. The read number counts nodes in the entire tree, not queries along one path.

For a read-k tree, the bound is $I(f) \log_2 k$. The dummy-prefix construction of Wan, Wright, and Wu attains equality when k is a power of two. The logarithmic potential is due to Shalev. A four-variable counterexample has covariance 5/4 and variance 3/4. A nine-variable counterexample has covariance 4863/4096 and variance 3007/4096, with every query relevant. Both refute the coefficient-one logarithmic variance proposal for a given tree.

Scope: Covariance inequalities, an equality family, and exact finite witnesses. No Fourier-entropy theorem is formalized.

Manuscript Prior work Formal proof and reproduction.

3.3 Fair division

3.3.1 Ordered-instance allocation transfer with exact extra-copy preservation

A set-valued ordered-instance transfer preserving utility improvement, the exact extra-copy count, per-good caps and coverage. The transfer theorem is formalized. Existence of a fair allocation before transfer is outside the result; external source comparison is provisional.

Manuscript Reproduction and source context.

3.4 Proof complexity

3.4.1 Unbounded coalesced-mass certificates for resolution trees

A coalesced dyadic clause-mass functional has exact supremum n+1 on the stated unrestricted resolution-tree class and is unbounded across dimensions. The internally defined resolution-tree functional and proof class are formalized. No unconditional identification with an external conjecture or a regular proof class is claimed.

Manuscript Reproduction and source context.

4 Mathematical physics

4.1 Stochastic thermodynamics

4.1.1 A three-state thermal counterexample to a heat-current kurtosis conjecture

A three-state rate model with two positive-temperature baths satisfies local detailed balance. Every total off-diagonal rate is positive. The tilted generator records heat transferred to the cold bath. Its dominant eigenvalue branch is four times continuously differentiable near zero and has a larger real part than every distinct complex eigenvalue. The exact derivative ratio is

$$\frac{C_4}{C_2} = -\frac{3470947982622836}{176627429595846691} < 0,$$

where C_n is the nth derivative at zero. This refutes Eq. (19) of Ptaszyński, *Physical Review E* **106**, 024119 (2022), in the paper's finite thermal spectral formulation. The third derivative is positive, so the witness does not refute the separate skewness conjecture.

Scope: Finite rate matrices, local detailed balance, complex eigenvalues, and derivatives of a local dominant branch. The stochastic long-time theorem that identifies the branch with a cumulant generating function is not formalized. No minimal-state or empirical validation claim is made.

Manuscript Prior work Formal proof and reproduction.

4.1.2 A sharp universal skewness bound for unicyclic currents

The net marked-edge current on every finite bidirectional Markov ring with at least three states, positive rates and unequal forward and reverse rate products satisfies $-1/16 \le \mathrm{C}_3/\mathrm{C}_1 \le 1$. The two-channel two-state case has the stronger interval [1/4,1]. This ratio is not standardized skewness.

Formal scope: The interval for actual tilted-matrix derivatives and a local branch is formalized. Identification with long-time stochastic cumulants and sharpness have conventional proofs. Lean does not construct a continuous-time sample-path probability space or a separate optimizing-sequence theorem.

Manuscript Proof and reproduction.

4.2 Quantum information

4.2.1 Universal entanglement embezzlement

A target-independent catalyst of local dimension 2^{4L} gives root infidelity at most $125m/L^2$ for every Schmidt probability vector on m entries and $L \ge 1$. Zero entries are permitted. The proof constructs local basis permutations and gives their finite complex Hilbert-space interpretation. It does not separately formalize arbitrary changes to an external Schmidt basis, gate efficiency, or catalyst reuse.

Manuscript Formal proof and reproduction.

4.2.2 The four-qubit Higuchi–Sudbery entropy bound

For every normalized pure state of four qubits, the average von Neumann entropy of the three balanced two-qubit cuts is at most $\log 2 + \frac{1}{2}\log 3$. The Higuchi–Sudbery state attains this value. The formal proof uses all sixteen complex amplitudes, the actual reduced matrices and their eigenvalues, an exact sum-of-squares identity, and an explicit attaining state. It does not classify all maximizers or cover mixed states.

Manuscript, Lean proof, and reproduction commands.

4.2.3 A pure-state matched-basis information inequality violation in dimension five

A dimension-five pure-state matched-basis information inequality violation, with a conventional all-prime companion. The dimension-five identical-local-basis theorem is formalized. The prime-family extension is conventional; arbitrary or independently optimized basis choices are outside the result.

Manuscript Reproduction and source context.

4.2.4 Exact three-setting correlation bounds at fixed steering centre

The exact least upper bound of the optimized three-setting squared correlation value is determined for every two-qubit steering-centre length. The threshold for a value greater than one is $c < 2/3$. Attainment, nonattainment and the singular marginal boundary are included.

Formal scope: Actual two-qubit density matrices, canonical filters, Pauli correlations, the exact least upper bound, attainment and nonattainment, and fully entangled fraction are formalized. The steering-ellipsoid and optimized CJWR measurement interpretations use cited external correspondences; general steerability is outside the result.

Manuscript Proof and reproduction.

4.2.5 The three-state root-fidelity entropy bound

The entropy of the weighted three-by-three root-fidelity matrix bounds the Holevo quantity of every three-state ensemble in arbitrary finite dimension. The result includes mixed and singular states and zero probabilities.

Formal scope: The unrestricted three-state ensemble theorem is formalized for actual positive semidefinite complex matrices of trace one, nonnegative weights summing to one, spectral entropy and nuclear-norm root fidelity. Credited QICLean foundations and prior-work adapters retain their attribution.

Manuscript Proof and reproduction.

4.2.6 Exact conditional contraction for Pauli channels and depolarization in every dimension

The conditional hockey-stick contraction coefficient is determined for every stochastic Pauli qubit channel and for depolarization throughout its physical parameter interval in every dimension $d \geq 2$. The formulas include all boundary cases and every parameter $\gamma \geq 1$, with arbitrary finite quantum references.

Formal scope: Actual channel constructors, unrestricted upper bounds and matching density-state supremum witnesses are formalized. Unital-qubit normal-form existence is cited. Erasure-simulation and ensemble-guessing consequences retain conventional proofs.

Manuscript Proof and reproduction.

4.2.7 Strong minimum-rank multiplicativity from unextendible product constraints

A complex matrix subspace annihilated by an unextendible family of product functionals doubles minimum rank under tensor products with any partner. For the Tiles bound entangled state ρ, the Schmidt number of $\rho^{\otimes N}$ is exactly 2^N for every $N \geq 1$, extending the known two-copy case. The general theorem and the all-copy formula are formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

4.2.8 No universal finite collapse of the upper PPT entanglement-cost hierarchy

For every $q \geq 1$ there is a full-rank Bell-diagonal state in local dimension $2^{(2q+4)4^q}$ with $\kappa_{q+1}(\rho) < \kappa_q(\rho)$ in the upper hierarchy of Lami, Mele and Regula, so no fixed finite upper level is exact for all states. Brinster et al. had proved the corresponding statement for the lower hierarchy. The strict separation for the actual semidefinite programs is formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

5 Manuscripts and proof packages

The repository contains the manuscripts, source files and supporting proof packages. The README highlights sixteen standout results: the upper PPT hierarchy, matrix-pencil rank distance, depth-one Artin–Schreier towers, nonnegative rank of Kronecker squares, Higuchi–Sudbery, universal embezzlement, asymmetric Tutte, power-rank, the ADMM multiplier-step interval, the small-coefficient Keller–Klein bound, subcritical Nesterov flows and iterates, steering bounds, the strong-convexity gradient bound, unicyclic skewness, three-state root fidelity and exact conditional quantum contraction. Their manuscripts state the autonomous AI research disclosure and exact verification scope. The catalogue links all retained results and reproduction instructions.

6 Verification

Lean checks the proofs against their formal statements. In the shared Lean 4.35.0 library, separate Challenge modules state the intended claims. Comparator checks these against their proved Solutions. The verification record identifies the completed runs of Lean, NanoDa, and con-ron for those seven families. The standalone study packages have their own pinned Lean environments and stated proof boundaries. Source-integrity checks are distinguished from completed compilation records. Their audit logs state the permitted axioms. Exact arithmetic programs check the accompanying finite certificates. The shared Comparator results do not cover those packages. Independent exact and numerical programs test finite certificates and examples. These finite computations do not replace the general proofs.